

Name: Solutions

These questions assess your understanding of the material from Chapters 11, 12 of OpenStax College Physics. Please write your name on the sheets of paper that you use and show all your work.

1. The aorta pumps blood from the heart to the rest of the body. Calculate the average speed of the blood in Abigail's aorta if the flow rate of her blood is 11.4 L/min. Abigail's aorta has a radius of 11 mm.

find $\bar{v}$

$$Q = \frac{11.4 \text{ L}}{\text{min}} \left| \frac{1 \text{ m}^3}{1000 \text{ L}} \right| \left| \frac{\text{min}}{60 \text{ s}} \right| = 1.9 \times 10^{-4} \text{ m}^3/\text{s}$$

ANSWER: 49.98 cm/s

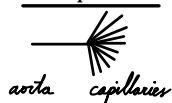
$$Q = A \bar{v}$$

$$\bar{v} = \frac{Q}{A}$$

$$\bar{v} = \frac{Q}{\pi r^2}$$

$$\bar{v} = \frac{1.9 \times 10^{-4} \frac{\text{m}^3}{\text{s}}}{\pi (11 \times 10^{-3} \text{ m})^2} = \underline{0.4998 \frac{\text{m}}{\text{s}}}$$

2. The aorta pumps blood from the heart to the rest of the body. The flow rate of blood through Leah's aorta is 12.0 L/min and Leah's aorta has a radius of 12 mm. Recall that blood flows through smaller blood vessels known as capillaries, which have an average diameter of 8.1 μm . The speed of blood in the capillaries is about 0.3 mm/s. Calculate the number of capillaries in Leah's blood circulatory system.



Derive an equation for n_c

$$Q_a = Q_c$$

$$Q_a = n_c A_c \bar{v}_c$$

$$n_c = \frac{Q_a}{A_c \bar{v}_c} = \frac{Q_a}{\pi r_c^2 \bar{v}_c}$$

$$n_c = \frac{Q_a}{\pi \left(\frac{d_c}{2}\right)^2 \bar{v}_c}$$

$$n_c = \frac{4 Q_a}{\pi d_c^2 \bar{v}_c}$$

get everything in SI units

$$Q_a = \frac{12 \text{ L}}{\text{min}} \left| \frac{1 \text{ m}^3}{1000 \text{ L}} \right| \left| \frac{\text{min}}{60 \text{ s}} \right| = 2 \times 10^{-4} \text{ m}^3/\text{s}$$

$$d_c = 8.1 \times 10^{-6} \text{ m}$$

$$\bar{v}_c = 3 \times 10^{-4} \frac{\text{m}}{\text{s}}$$

find n_c

ANSWER: 1.29×10^{10} capillaries

solve

$$n_c = \frac{4 (2 \times 10^{-4} \frac{\text{m}^3}{\text{s}})}{\pi (8.1 \times 10^{-6} \text{ m})^2 (3 \times 10^{-4} \frac{\text{m}}{\text{s}})} = \underline{1.29 \times 10^{10} \text{ capillaries}}$$

3. A particular hydraulic system contains two fluid-filled cylinders which are connected by a tube. Cylinder 1 has a diameter of 2.4 cm and is capped with piston P_1 . Cylinder 2 has a diameter of 9.5 cm and is capped with piston P_2 . A force F_1 , equal to 50 N, is exerted on P_1 . What is the magnitude of the resulting force exerted by P_2 ?

Derive an equation for F_2

$$P_1 = P_2 \quad (\text{Pascal's principle})$$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$F_2 = F_1 \left(\frac{A_2}{A_1} \right)$$

$$F_2 = F_1 \left(\frac{\pi r_2^2}{\pi r_1^2} \right)$$

$$F_2 = F_1 \left(\frac{\left(\frac{d_2}{2}\right)^2}{\left(\frac{d_1}{2}\right)^2} \right)$$

$$F_2 = F_1 \left(\frac{d_2^2}{d_1^2} \right)$$

$$F_2 = F_1 \left(\frac{d_2}{d_1} \right)^2$$

find F_2

ANSWER: 783.42 N

Solve for F_2

$$F_2 = F_1 \left(\frac{d_2}{d_1} \right)^2$$

$$F_2 = 50 \text{ N} \left(\frac{9.5 \text{ cm}}{2.4 \text{ cm}} \right)^2 = \underline{783.42 \text{ N}}$$

Notice that d_1 & d_2 can remain in cm because the units cancel in the quotient

(due to the weight of water)

4. Paul's doctor performs a special test and tells him that a 3.3-N force can rupture his eardrum. If his eardrum has an area of 1.0 cm², at what depth in freshwater would his eardrum rupture, assuming the gauge pressure in the middle ear is zero and he does nothing to alleviate the pressure? *find h*

ANSWER: 3.37 m

$$h \uparrow \cdot P_1 = P_{atm}$$

$$h \downarrow \cdot P_2 = P_{atm} + \rho g h$$

Derive an equation for h

$$\Delta P = P_2 - P_1 = \rho g h$$

$$\frac{F}{A} = \rho g h$$

$$h = \frac{F}{A \rho g}$$

solve for h

$$h = \frac{F}{A \rho g}$$

$$h = \frac{3.3 \text{ N}}{\left(1 \text{ cm}^2 \left| \frac{\text{m}^2}{100^2 \text{ cm}^2} \right| \right) \left(1000 \frac{\text{kg}}{\text{m}^3} \right) \left(9.8 \frac{\text{m}}{\text{s}^2} \right)} = \underline{3.37 \text{ m}}$$

5. Katherine is an astronaut working outside the International Space Station (ISS) where the atmospheric pressure is essentially zero. The pressure gauge on the astronaut's air tank reads $6.6 \times 10^6 \text{ Pa}$. What force does the air inside the tank exert on the flat end of the cylindrical tank, a disk that is 0.1 m in diameter? *find F*

ANSWER: $5.2 \times 10^4 \text{ N}$

$$P = \frac{F}{A}$$

$$F = A P$$

$$F = \pi r^2 P$$

$$F = \pi \left(\frac{d}{2}\right)^2 P$$

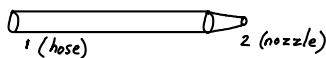
$$F = \frac{\pi d^2 P}{4}$$

$$F = \frac{\pi (0.1 \text{ m})^2 (6.6 \times 10^6 \text{ Pa})}{4} = \underline{5.2 \times 10^4 \text{ N}}$$

derive an equation for F & solve

6. Grace turns a hose faucet on and after 8.5 seconds, sees water emerge from her 5-m-long garden hose of diameter 2.9 cm. Grace then turns it off momentarily to attach a nozzle of diameter 6.7 mm to the end, then turns the water back on. She then uses the hose to fill up a 2-L bucket. What is the speed of the water that leaves the nozzle?

ANSWER: 11.02 m/s



find $\bar{v}_2$

supplementary equations

$$\bar{v}_1 = \frac{\Delta x_1}{t}$$

$$A_1 = \frac{\pi d_1^2}{4}$$

$$A_2 = \frac{\pi d_2^2}{4}$$

(see work in #5 for area as a function of diameter)

$$\therefore \frac{A_1}{A_2} = \frac{\frac{\pi d_1^2}{4}}{\frac{\pi d_2^2}{4}} = \left(\frac{d_1}{d_2}\right)^2$$

derive an equation for $\bar{v}_2$

$$Q_1 = Q_2$$

$$A_1 \bar{v}_1 = A_2 \bar{v}_2$$

$$\bar{v}_2 = \bar{v}_1 \left(\frac{A_1}{A_2}\right)$$

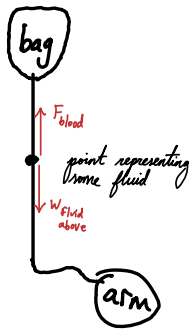
$$\bar{v}_2 = \frac{\Delta x_1}{t} \left(\frac{d_1}{d_2}\right)^2$$

solve for $\bar{v}_2$

$$\bar{v}_2 = \frac{\Delta x_1}{t} \left(\frac{d_1}{d_2}\right)^2$$

$$\bar{v}_2 = \frac{5 \text{ m}}{8.5 \text{ s}} \left(\frac{29 \text{ mm}}{6.7 \text{ mm}}\right)^2 = \underline{11.02 \frac{\text{m}}{\text{s}}}$$

7. Intravenous (IV) infusions are usually made with the help of the gravitational force. Assuming that the density of the fluid being administered is 1.00 g/mL , at what height should the IV bag be placed above the entry point so that the fluid only just enters the vein if the blood pressure in the vein is 7.1 mmHg above atmospheric pressure? Recall that $1.0 \text{ mmHg} = 133.32 \text{ Pa}$.



derive an equation for h

no flow $\therefore$ static

i.e. $F_{\text{blood}} = W_{\text{fluid above}}$

$$P_{\text{blood}} A = m_{\text{fluid}} g$$

$$P_{\text{blood}} A = \rho_{\text{fluid}} V g$$

$$P_{\text{blood}} A = \rho_{\text{fluid}} A h g$$

$$P_{\text{blood}} = \rho_{\text{fluid}} g h$$

$$h = \frac{P_{\text{blood}}}{\rho_{\text{fluid}} g}$$

ANSWER: 9.66 cm

solve for h

$$h = \frac{P_{\text{blood}}}{\rho_{\text{fluid}} g}$$

$$h = \frac{7.1 \text{ mmHg} \left| \frac{133.32 \text{ Pa}}{1 \text{ mmHg}} \right|}{\left(1000 \frac{\text{kg}}{\text{m}^3} \right) 9.8 \frac{\text{m}}{\text{s}^2}} = \underline{0.0966 \text{ m}}$$

8. Calculate the maximum buoyant force that water could exert on a ship with a mass of $1.1 \times 10^8 \text{ kg}$ if the ship displaces $4.6 \times 10^6 \text{ m}^3$ of water.

ANSWER: $4.51 \times 10^{10} \text{ N}$

Buoyant force F_B

$$F_B = W_{\text{fluid displaced}}$$

$$F_B = m_F g$$

$$F_B = \rho_F V g$$

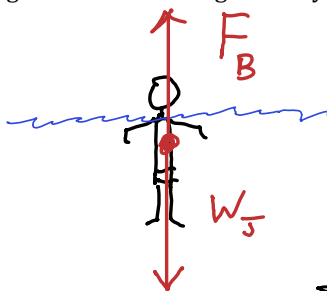
Solve for F_B

$$F_B = \rho_F V g$$

$$F_B = \left(1000 \frac{\text{kg}}{\text{m}^3} \right) \left(4.6 \times 10^6 \text{ m}^3 \right) \left(9.8 \frac{\text{m}}{\text{s}^2} \right) = \underline{4.51 \times 10^{10} \text{ N}}$$

9. Suppose that John floats in freshwater with 95.4% of his volume submerged when his lungs are full of air. His mass is 71 kg . What is his average density? find ρ_{John}

ANSWER: 954 kg/m^3



$\Sigma \vec{F}_y = m \vec{a}_y = 0$ when 95.4% of John's volume is submerged (equilibrium)

$$\Sigma \vec{F}_y = m \vec{a}_y = 0$$

$$\vec{F}_B + \vec{W}_J = 0$$

$$F_B - W_J = 0$$

$$F_B = W_J$$

$$m_F g = m_J g$$

$$m_F = m_J$$

$$\rho_F V_F = m_J$$

$$\rho_F (0.954 V_J) = m_J$$

$$0.954 \rho_F = \frac{m_J}{V_J}$$

$$0.954 \rho_F = \rho_J$$

$$\rho_J = 0.954 \left(1000 \frac{\text{kg}}{\text{m}^3} \right) = \underline{954 \frac{\text{kg}}{\text{m}^3}}$$

10. Jonathan creates a soap bubble and measures its diameter to be $296 \mu\text{m}$. Calculate the absolute pressure inside of the bubble, given that the surface tension of typical soapy water is 0.0370 N/m . Report your answer in mmHg, recalling that 133.32 Pa is 1 mmHg .

find P in bubble

ANSWER: 767.5 mmHg

derive an equation for P

$$P = P_g + P_{\text{atm}}$$

$$P = \frac{4\gamma}{r} + P_{\text{atm}}$$

$$P = \frac{8\gamma}{d} + P_{\text{atm}}$$

Solve for P in mmHg

$$P = \frac{8\gamma}{d} + P_{\text{atm}}$$

$$P = \frac{8(0.0370 \frac{\text{N}}{\text{m}})}{296 \times 10^{-6} \text{ m}} + 101325 \text{ Pa} = 102325 \text{ Pa}$$

$$102325 \text{ Pa} \left| \frac{1 \text{ mmHg}}{133.32 \text{ Pa}} \right| = \underline{767.5 \text{ mmHg}}$$

11. Calculate the depth below the surface of water at which the pressure due to the weight of the water equals 0.13 atm .

ANSWER: 1.34 m

h ↓

- $P = P_{\text{atm}}$
- $P = P_g + P_{\text{atm}}$

Derive an equation for h

$$P_g = \rho g h$$

$$h = \frac{P_g}{\rho g}$$

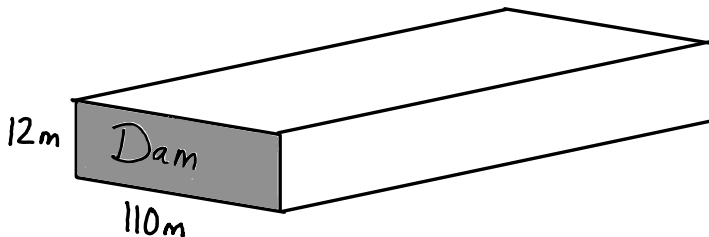
Solve for h

$$h = \frac{P_g}{\rho g}$$

$$h = \frac{(0.13 \text{ atm}) \left| \frac{101325 \text{ Pa}}{1 \text{ atm}} \right|}{(1000 \frac{\text{kg}}{\text{m}^3}) (9.8 \frac{\text{m}}{\text{s}^2})} = \underline{1.34 \text{ m}}$$

12. A particular dam is 110-m wide and contains water that is 12-m deep. What is the average pressure on the dam due to the water?

ANSWER: 58.8 kPa



- water at the top exerts the most pressure
- water at the bottom exerts no pressure
- average pressure is at half the depth (6 m)

$$P_g = \rho g h$$

$$P_g = 1000 \frac{\text{kg}}{\text{m}^3} (9.8 \frac{\text{m}}{\text{s}^2}) \left(\frac{12 \text{ m}}{2} \right) = \underline{58800 \text{ Pa}}$$